Statistical Inference: Mental Health & Social Determinants (BRFSS 2024)

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```
library(dplyr)
library(ggplot2)
library(tidyr)
library(forcats)
library(knitr)
library(broom)
library(scales)
```

```
# Load cleaned CSV dataset
brfss1 <- read.csv("brfss_names.csv")
cat("Dataset:", nrow(brfss1), "rows,", ncol(brfss1), "cols\n")
```

```
## Dataset: 200326 rows, 25 cols
```

```
# Re-apply factor levels since CSV reloads as character
inf_df <- brfss1 %>%
  mutate(
    MENTHLTH = as.numeric(MENTHLTH),
    SEX      = factor(trimws(X_SEX),
                      levels = c("Male","Female")),
    AGE      = factor(trimws(X_AGEG5YR),
                      levels = c("18-24","25-29","30-34","35-39","40-44",
```

```
                                "45-49","50-54","55-59","60-64","65-69",
                                "70-74","75-79","80+")),
    INCOME   = factor(trimws(X_INCOMG1),
                      levels = c("<$15k","$15-25k","$25-35k","$35-50k",
                                 "$50k-$100k","$100k-$200k",">$200k")),
    SMOKER   = factor(trimws(X_SMOKER3),
                      levels = c("Everyday smoker","Someday smoker",
                                 "Former smoker","Never smoked")),
    LONELY   = factor(trimws(SDLONELY),
                      levels = c("Always","Usually","Sometimes","Rarely","Never")),
    EMOSUP   = factor(trimws(EMTSUPRT),
                      levels = c("Always","Usually","Sometimes","Rarely","Never")),
    LIFESAT  = factor(trimws(LSATISFY),
                      levels = c("Very satisfied","Satisfied",
                                 "Dissatisfied","Very dissatisfied")),
    DEPRESS  = factor(trimws(ADDEPEV3), levels = c("Yes","No")),
    GENHLTH  = factor(trimws(GENHLTH),
                      levels = c("Excellent","Very good","Good","Fair","Poor")),
    EXERCISE = factor(trimws(EXERANY2), levels = c("Yes","No")),
    DRINK    = factor(trimws(DRNKANY6), levels = c("Yes","No"))
  ) %>%
  filter(!is.na(MENTHLTH))

cat("Analysis dataset:", nrow(inf_df), "complete rows\n")
```

```
## Analysis dataset: 200326 complete rows
```

---

1. Descriptive Statistics

```
desc <- inf_df %>%
  summarise(
    N      = n(),
    Mean   = round(mean(MENTHLTH, na.rm = TRUE), 2),
    SD     = round(sd(MENTHLTH,   na.rm = TRUE), 2),
    Median = median(MENTHLTH, na.rm = TRUE),
    IQR    = IQR(MENTHLTH,    na.rm = TRUE),
    Min    = min(MENTHLTH,    na.rm = TRUE),
    Max    = max(MENTHLTH,    na.rm = TRUE)
  )
kable(desc, caption = "Descriptive Statistics: Poor Mental Health Days (Past 30 Days)")
```

Table 1: Descriptive Statistics: Poor Mental Health Days (Past 30 Days)

N	Mean	SD	Median	IQR	Min	Max
200326	4.26	8.23	0	4	0	30

```
group_stats <- function(df, grp_var, grp_label) {
  df %>%
    filter(!is.na(.data[[grp_var]])) %>%
```

```
    group_by(Group = .data[[grp_var]]) %>%
    summarise(N    = n(),
              Mean = round(mean(MENTHLTH, na.rm = TRUE), 2),
              SD   = round(sd(MENTHLTH,   na.rm = TRUE), 2),
              .groups = "drop") %>%
    mutate(Variable = grp_label) %>%
    select(Variable, Group, N, Mean, SD)
}

bind_rows(
  group_stats(inf_df, "SEX",      "Sex"),
  group_stats(inf_df, "DEPRESS", "Depression Diagnosis"),
  group_stats(inf_df, "LONELY",  "Loneliness"),
  group_stats(inf_df, "LIFESAT", "Life Satisfaction")
) %>%
  kable(caption = "Mean Mental Health Days by Key Social Groups")
```

Table 2: Mean Mental Health Days by Key Social Groups

Variable	Group	N	Mean	SD
Sex	Male	94603	3.62	7.75
Sex	Female	105723	4.83	8.58
Depression Diagnosis	Yes	41781	10.98	11.10
Depression Diagnosis	No	157608	2.45	6.11
Loneliness	Always	4903	16.12	12.94
Loneliness	Usually	7240	13.64	11.54
Loneliness	Sometimes	46356	7.18	9.53
Loneliness	Rarely	64418	3.10	6.49
Loneliness	Never	75964	1.78	5.67
Life Satisfaction	Very satisfied	89647	1.71	5.04
Life Satisfaction	Satisfied	98163	5.12	8.51
Life Satisfaction	Dissatisfied	8569	15.60	11.71
Life Satisfaction	Very dissatisfied	2371	20.10	12.12

Interpretation: This table reveals important patterns – respondents with depression, high loneliness, or low life satisfaction report substantially more poor mental health days than their counterparts.

---

2. Hypothesis Testing

2.1 t-Test: Mental Health Days by Sex

H0: Mean mental health days are equal for Males and Females **H1:** Mean mental health days differ between Males and Females

```
male   <- inf_df$MENTHLTH[inf_df$SEX == "Male"]
female <- inf_df$MENTHLTH[inf_df$SEX == "Female"]

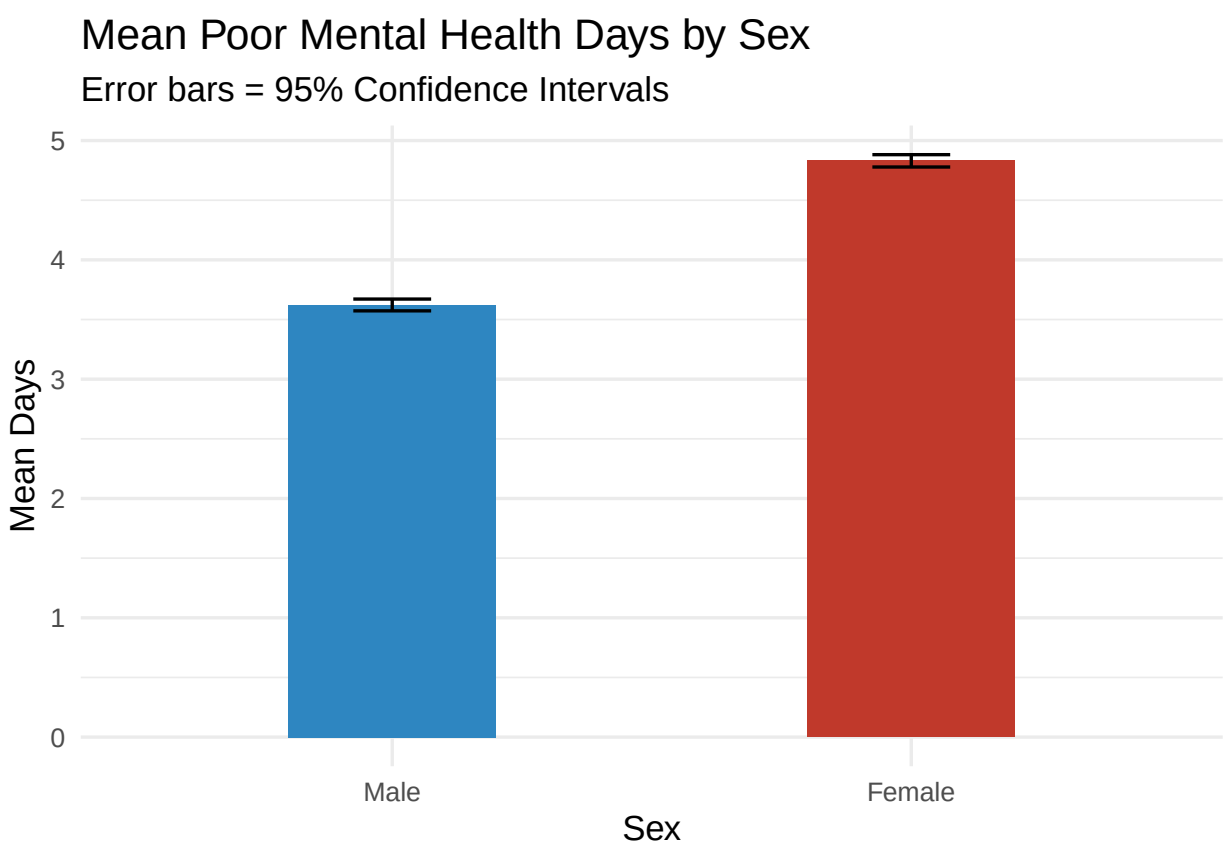
t_res <- t.test(female, male, alternative = "two.sided")
tidy(t_res) %>%
  select(estimate1, estimate2, statistic, p.value, conf.low, conf.high) %>%
  rename("Mean (Female)" = estimate1, "Mean (Male)" = estimate2,
```

```
        "t-stat" = statistic, "p-value" = p.value,
        "CI Low" = conf.low, "CI High" = conf.high) %>%
  mutate(across(where(is.numeric), ~ round(., 4))) %>%
  kable(caption = "Welch t-Test: Mental Health Days by Sex")
```

Table 3: Welch t-Test: Mental Health Days by Sex

Mean (Female)	Mean (Male)	t-stat	p-value	CI Low	CI High
4.8301	3.6222	33.0894	0	1.1364	1.2795

```
inf_df %>%
  filter(!is.na(SEX)) %>%
  group_by(SEX) %>%
  summarise(mean  = mean(MENTHLTH, na.rm = TRUE),
            se    = sd(MENTHLTH, na.rm = TRUE) / sqrt(n()),
            .groups = "drop") %>%
  mutate(lower = mean - 1.96 * se, upper = mean + 1.96 * se) %>%
  ggplot(aes(x = SEX, y = mean, fill = SEX)) +
  geom_col(width = 0.4, show.legend = FALSE) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  scale_fill_manual(values = c("Male" = "#2E86C1", "Female" = "#C0392B")) +
  labs(title    = "Mean Poor Mental Health Days by Sex",
       subtitle = "Error bars = 95% Confidence Intervals",
       x = "Sex", y = "Mean Days") +
  theme_minimal(base_size = 13)
```



Interpretation: If p-value < 0.05, females report significantly more poor mental health days than males, even after accounting for sampling variability.

2.2 t-Test: Mental Health Days by Depression Diagnosis

H0: Mean mental health days are equal regardless of depression diagnosis **H1:** Those with depression have more poor mental health days

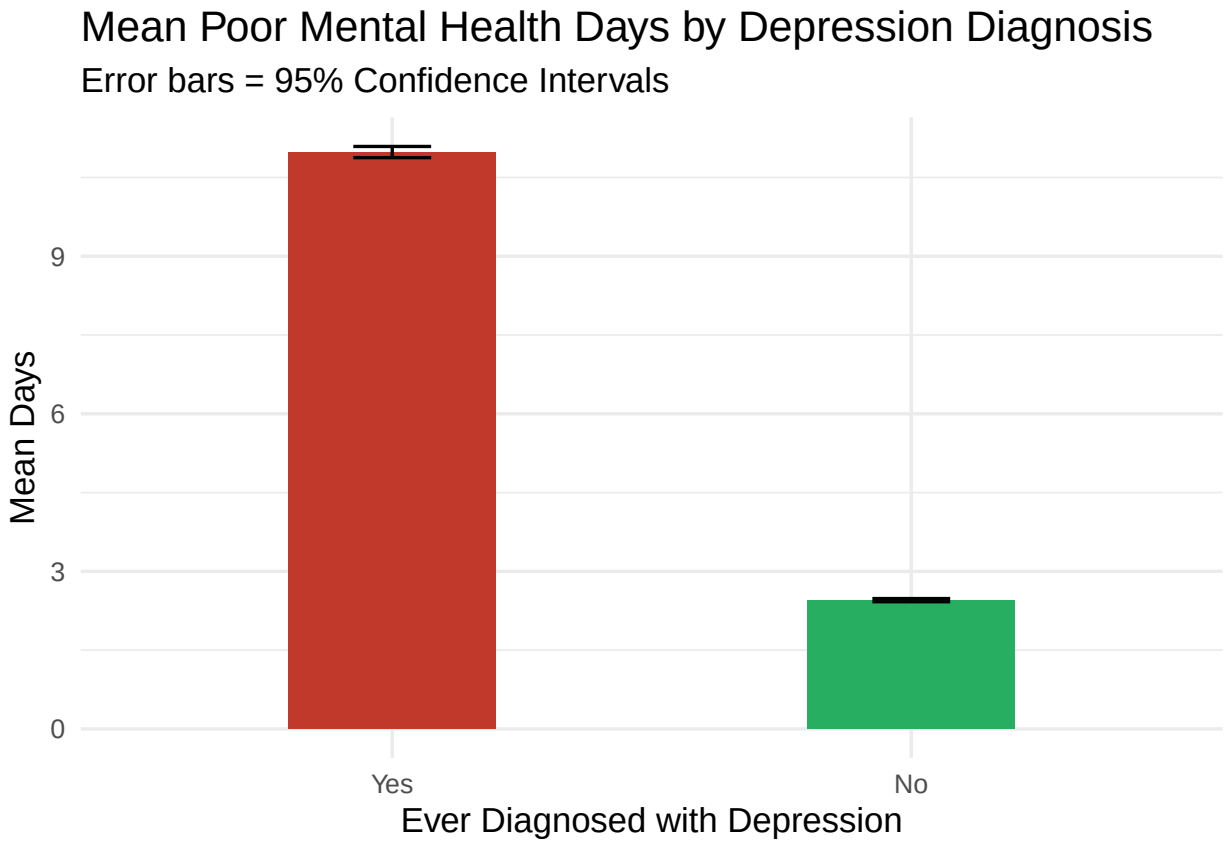
```
dep_yes <- inf_df$MENTHLTH[inf_df$DEPRESS == "Yes"]
dep_no  <- inf_df$MENTHLTH[inf_df$DEPRESS == "No"]

t_dep <- t.test(dep_yes, dep_no, alternative = "greater")
tidy(t_dep) %>%
  select(estimate1, estimate2, statistic, p.value, conf.low, conf.high) %>%
  rename("Mean (Depressed)" = estimate1, "Mean (Not Depressed)" = estimate2,
         "t-stat" = statistic, "p-value" = p.value,
         "CI Low" = conf.low, "CI High" = conf.high) %>%
  mutate(across(where(is.numeric), ~ round(., 4))) %>%
  kable(caption = "t-Test: Mental Health Days by Depression Diagnosis")
```

Table 4: t-Test: Mental Health Days by Depression Diagnosis

Mean (Depressed)	Mean (Not Depressed)	t-stat	p-value	CI Low	CI High
10.9843	2.4515	151.1815	0	8.4399	Inf

```
inf_df %>%
  filter(!is.na(DEPRESS)) %>%
  group_by(DEPRESS) %>%
  summarise(mean  = mean(MENTHLTH, na.rm = TRUE),
            se    = sd(MENTHLTH, na.rm = TRUE) / sqrt(n()),
            .groups = "drop") %>%
  mutate(lower = mean - 1.96 * se, upper = mean + 1.96 * se) %>%
  ggplot(aes(x = DEPRESS, y = mean, fill = DEPRESS)) +
  geom_col(width = 0.4, show.legend = FALSE) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  scale_fill_manual(values = c("Yes" = "#C0392B", "No" = "#27AE60")) +
  labs(title    = "Mean Poor Mental Health Days by Depression Diagnosis",
       subtitle = "Error bars = 95% Confidence Intervals",
       x = "Ever Diagnosed with Depression", y = "Mean Days") +
  theme_minimal(base_size = 13)
```



Interpretation: This is the most clinically important test. A significant result confirms that depression diagnosis is strongly associated with more poor mental health days – validating the survey variable's construct validity.

2.3 ANOVA: Mental Health Days by Loneliness

H0: Mean mental health days are equal across all loneliness levels **H1:** At least one loneliness group differs

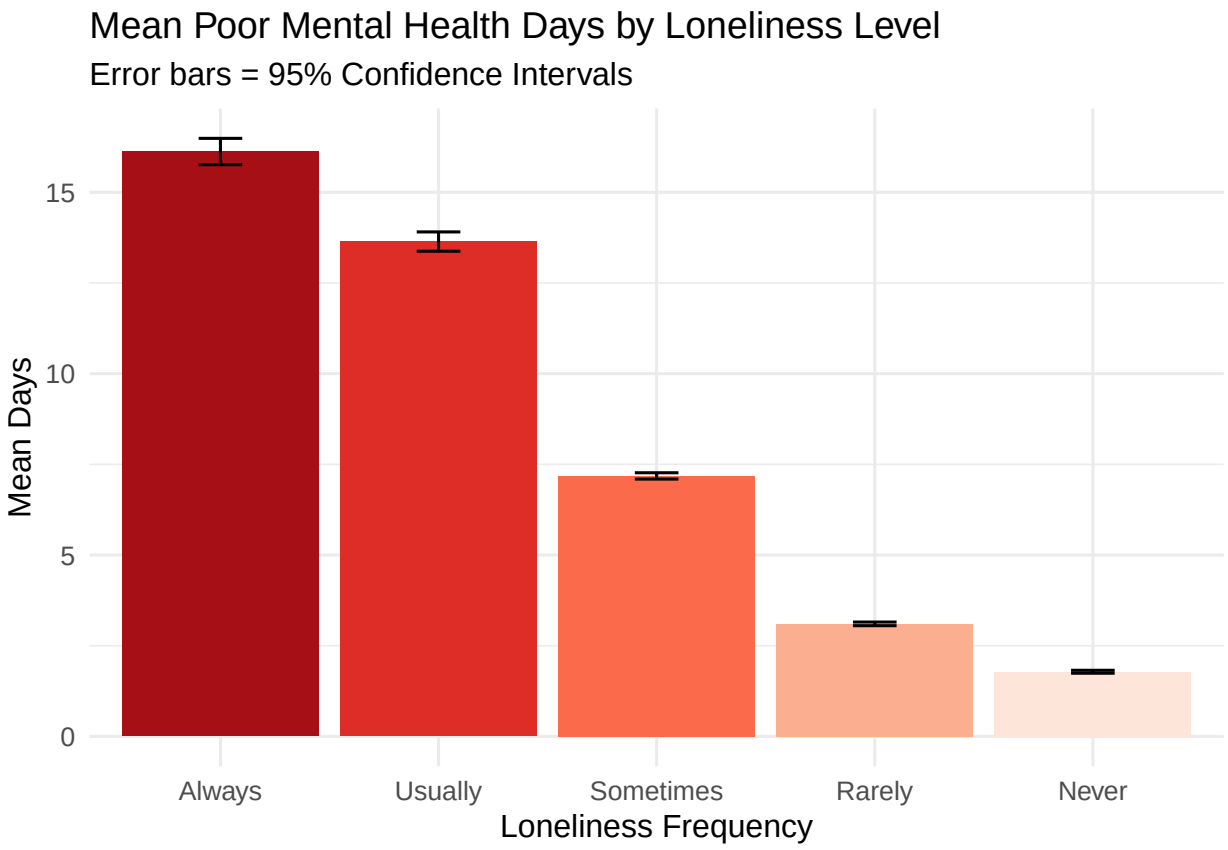
```
anova_lonely <- aov(MENTHLTH ~ LONELY, data = inf_df)
tidy(anova_lonely) %>%
  mutate(across(where(is.numeric), ~ round(., 4))) %>%
  kable(caption = "One-Way ANOVA: Mental Health Days by Loneliness Level")
```

Table 5: One-Way ANOVA: Mental Health Days by Loneliness Level

term	df	sumsq	meansq	statistic	p.value
LONELY	4	2274805	568701.3167	10150.01	0
Residuals	198876	11142951	56.0296	NA	NA

```
inf_df %>%
  filter(!is.na(LONELY)) %>%
  group_by(LONELY) %>%
  summarise(mean  = mean(MENTHLTH, na.rm = TRUE),
            se    = sd(MENTHLTH, na.rm = TRUE) / sqrt(n()),
```

```
            .groups = "drop") %>%
  mutate(lower = mean - 1.96 * se, upper = mean + 1.96 * se) %>%
  ggplot(aes(x = LONELY, y = mean, fill = LONELY)) +
  geom_col(show.legend = FALSE) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.2) +
  scale_fill_brewer(palette = "Reds", direction = -1) +
  labs(title    = "Mean Poor Mental Health Days by Loneliness Level",
       subtitle = "Error bars = 95% Confidence Intervals",
       x = "Loneliness Frequency", y = "Mean Days") +
  theme_minimal(base_size = 12)
```



Interpretation: Loneliness is a powerful social determinant of mental health. Those who feel lonely "Always" are expected to report dramatically more poor mental health days than those who "Never" feel lonely.

2.4 ANOVA: Mental Health Days by Life Satisfaction

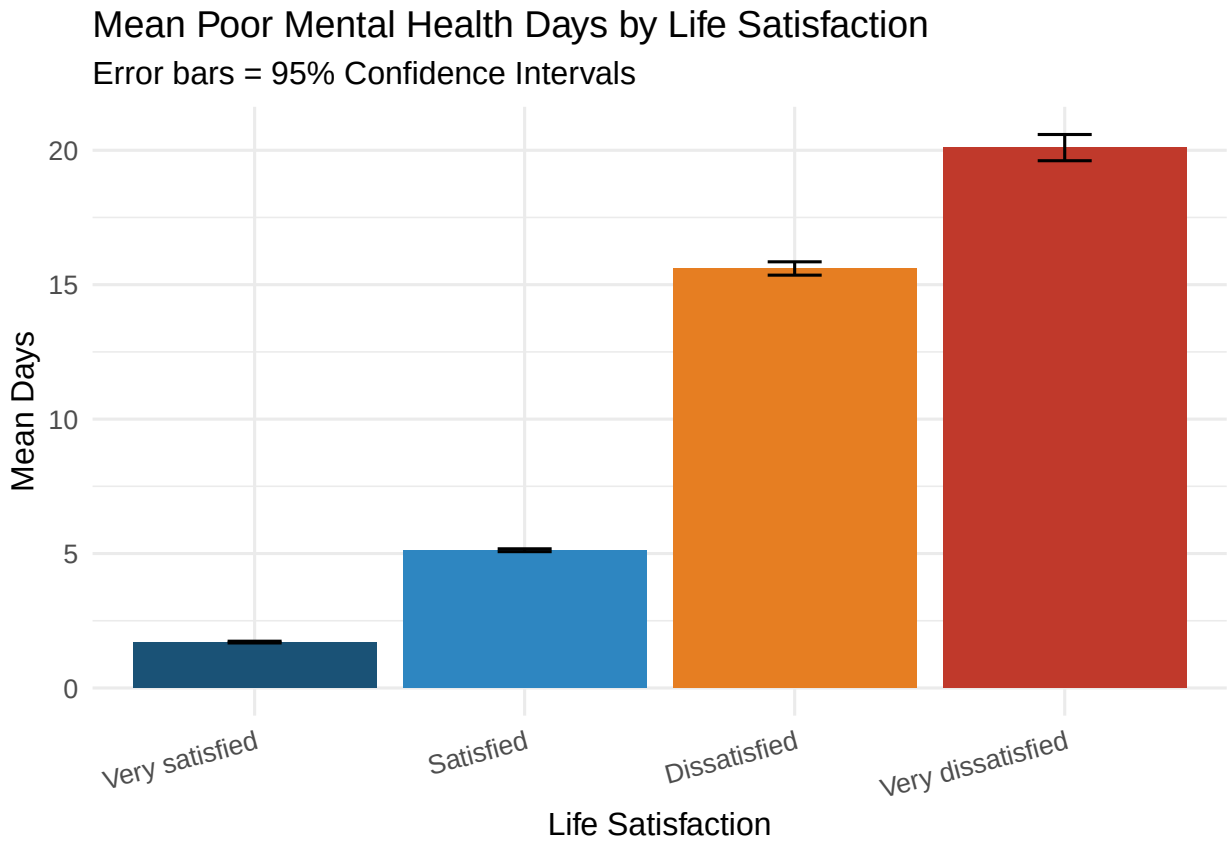
H0: Mean mental health days are equal across life satisfaction levels **H1:** At least one life satisfaction group differs

```
anova_life <- aov(MENTHLTH ~ LIFESAT, data = inf_df)
tidy(anova_life) %>%
  mutate(across(where(is.numeric), ~ round(., 4))) %>%
  kable(caption = "One-Way ANOVA: Mental Health Days by Life Satisfaction")
```

Table 6: One-Way ANOVA: Mental Health Days by Life Satisfaction

term	df	sumsq	meansq	statistic	p.value
LIFESAT	3	2354322	784773.9052	14299.63	0
Residuals	198746	10907319	54.8807	NA	NA

```
inf_df %>%
  filter(!is.na(LIFESAT)) %>%
  group_by(LIFESAT) %>%
  summarise(mean  = mean(MENTHLTH, na.rm = TRUE),
            se    = sd(MENTHLTH, na.rm = TRUE) / sqrt(n()),
            .groups = "drop") %>%
  mutate(lower = mean - 1.96 * se, upper = mean + 1.96 * se) %>%
  ggplot(aes(x = LIFESAT, y = mean, fill = LIFESAT)) +
  geom_col(show.legend = FALSE) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.2) +
  scale_fill_manual(values = c("Very satisfied"     = "#1A5276",
                               "Satisfied"          = "#2E86C1",
                               "Dissatisfied"       = "#E67E22",
                               "Very dissatisfied"  = "#C0392B")) +
  labs(title    = "Mean Poor Mental Health Days by Life Satisfaction",
       subtitle = "Error bars = 95% Confidence Intervals",
       x = "Life Satisfaction", y = "Mean Days") +
  theme_minimal(base_size = 12) +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))
```



Interpretation: A clear gradient is expected – those who are "Very dissatisfied" with life report substantially more poor mental health days than those who are "Very satisfied". This validates life satisfaction as a meaningful predictor of mental health.

2.5 ANOVA: Mental Health Days by Emotional Support

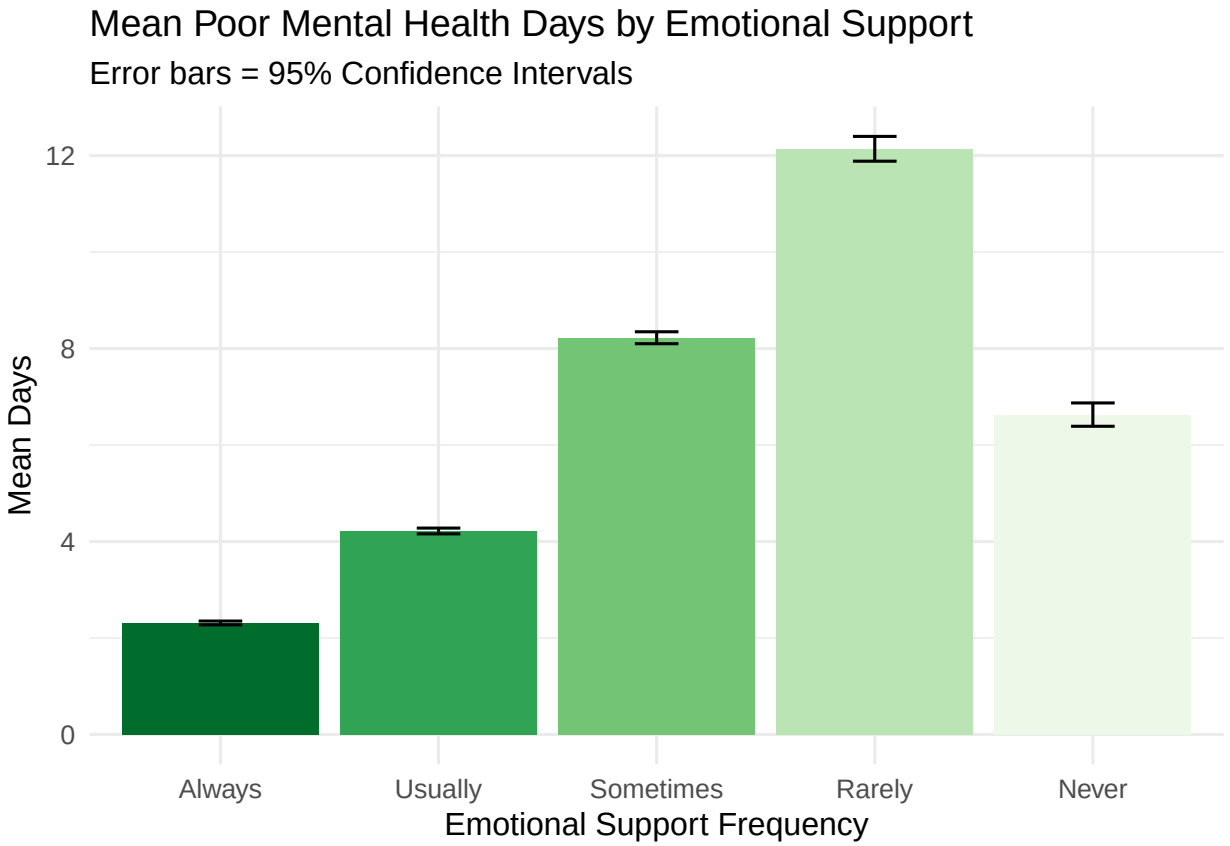
H0: Mean mental health days are equal across emotional support levels **H1:** At least one emotional support group differs

```
anova_emo <- aov(MENTHLTH ~ EMOSUP, data = inf_df)
tidy(anova_emo) %>%
  mutate(across(where(is.numeric), ~ round(., 4))) %>%
  kable(caption = "One-Way ANOVA: Mental Health Days by Emotional Support")
```

Table 7: One-Way ANOVA: Mental Health Days by Emotional Support

term	df	sumsq	meansq	statistic	p.value
EMOSUP	4	1346895	336723.669	5538.947	0
Residuals	197454	12003623	60.792	NA	NA

```
inf_df %>%
  filter(!is.na(EMOSUP)) %>%
  group_by(EMOSUP) %>%
  summarise(mean  = mean(MENTHLTH, na.rm = TRUE),
            se    = sd(MENTHLTH, na.rm = TRUE) / sqrt(n()),
            .groups = "drop") %>%
  mutate(lower = mean - 1.96 * se, upper = mean + 1.96 * se) %>%
  ggplot(aes(x = EMOSUP, y = mean, fill = EMOSUP)) +
  geom_col(show.legend = FALSE) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.2) +
  scale_fill_brewer(palette = "Greens", direction = -1) +
  labs(title    = "Mean Poor Mental Health Days by Emotional Support",
       subtitle = "Error bars = 95% Confidence Intervals",
       x = "Emotional Support Frequency", y = "Mean Days") +
  theme_minimal(base_size = 12)
```



Interpretation: Those who "Never" receive emotional support are expected to report the most poor mental health days, while those who "Always" receive support report the fewest. This highlights the protective role of social support on mental health.

2.6 Tukey HSD: Loneliness Pairwise Comparisons

```
tukey_lonely <- TukeyHSD(anova_lonely)
tidy(tukey_lonely) %>%
  filter(adj.p.value < 0.05) %>%
  arrange(adj.p.value) %>%
  head(10) %>%
  mutate(across(where(is.numeric), ~ round(., 4))) %>%
  kable(caption = "Tukey HSD: Significant Loneliness Pairwise Comparisons (p < 0.05)")
```

Table 8: Tukey HSD: Significant Loneliness Pairwise Comparisons ($p < 0.05$)

term	contrast	null.value	estimate	conf.low	conf.high	adj.p.value
LONELY	Usually-Always	0	-2.4818	-2.8594	-2.1041	0
LONELY	Sometimes-Always	0	-8.9420	-9.2486	-8.6354	0
LONELY	Rarely-Always	0	-13.0196	-13.3221	-12.7171	0
LONELY	Never-Always	0	-14.3396	-14.6405	-14.0388	0
LONELY	Sometimes-Usually	0	-6.4602	-6.7183	-6.2022	0
LONELY	Rarely-Usually	0	-10.5378	-10.7909	-10.2847	0
LONELY	Never-Usually	0	-11.8579	-12.1090	-11.6067	0

term	contrast	null.value	estimate	conf.low	conf.high	adj.p.value
LONELY	Rarely-Sometimes	0	-4.0776	-4.2019	-3.9532	0
LONELY	Never-Sometimes	0	-5.3976	-5.5180	-5.2773	0
LONELY	Never-Rarely	0	-1.3200	-1.4294	-1.2107	0

Interpretation: The Tukey HSD test identifies exactly which pairs of loneliness groups have significantly different mental health days. The contrast between "Always" and "Never" lonely is expected to show the largest and most significant gap.

---

3. Confidence Intervals

```
ci_overall <- inf_df %>%
  summarise(n    = n(),
            mean = mean(MENTHLTH, na.rm = TRUE),
            se   = sd(MENTHLTH, na.rm = TRUE) / sqrt(n()),
            lower_95 = mean - 1.96 * se,
            upper_95 = mean + 1.96 * se) %>%
  mutate(Group = "Overall") %>%
  select(Group, n, mean, lower_95, upper_95)

ci_sex <- inf_df %>%
  filter(!is.na(SEX)) %>%
  group_by(Group = SEX) %>%
  summarise(n    = n(),
            mean = mean(MENTHLTH, na.rm = TRUE),
            se   = sd(MENTHLTH, na.rm = TRUE) / sqrt(n()),
            lower_95 = mean - 1.96 * se,
            upper_95 = mean + 1.96 * se,
            .groups = "drop") %>%
  mutate(Group = as.character(Group)) %>%
  select(Group, n, mean, lower_95, upper_95)

ci_dep <- inf_df %>%
  filter(!is.na(DEPRESS)) %>%
  group_by(Group = DEPRESS) %>%
  summarise(n    = n(),
            mean = mean(MENTHLTH, na.rm = TRUE),
            se   = sd(MENTHLTH, na.rm = TRUE) / sqrt(n()),
            lower_95 = mean - 1.96 * se,
            upper_95 = mean + 1.96 * se,
            .groups = "drop") %>%
  mutate(Group = paste("Depression:", Group)) %>%
  select(Group, n, mean, lower_95, upper_95)

bind_rows(ci_overall, ci_sex, ci_dep) %>%
  mutate(across(c(mean, lower_95, upper_95), ~ round(., 3))) %>%
  kable(caption = "95% Confidence Intervals for Mean Mental Health Days")
```

Table 9: 95% Confidence Intervals for Mean Mental Health Days

Group	n	mean	lower_95	upper_95
Overall	200326	4.260	4.224	4.296
Male	94603	3.622	3.573	3.672
Female	105723	4.830	4.778	4.882
Depression: Yes	41781	10.984	10.878	11.091
Depression: No	157608	2.451	2.421	2.482

Interpretation: Non-overlapping confidence intervals between groups confirm statistically significant differences. The gap between depressed and non-depressed groups is expected to be the largest.

---

4. Correlation Analysis

```
cor_df <- inf_df %>%
  mutate(
    SEX_num    = as.numeric(SEX),
    AGE_num    = as.numeric(AGE),
    INCOME_num = as.numeric(INCOME),
    SMOKER_num = as.numeric(SMOKER),
    LONELY_num = as.numeric(LONELY),
    EMOSUP_num = as.numeric(EMOSUP),
    LIFESAT_num= as.numeric(LIFESAT),
    DEPRESS_num= as.numeric(DEPRESS)
  ) %>%
  select(MENTHLTH, LONELY_num, LIFESAT_num, DEPRESS_num,
         EMOSUP_num, SEX_num, AGE_num, INCOME_num, SMOKER_num)

cor_matrix <- cor(cor_df, method = "spearman", use = "complete.obs")
round(cor_matrix[,"MENTHLTH", drop = FALSE], 3) %>%
  kable(caption = "Spearman Correlation with MENTHLTH (Mental Health Days)")
```

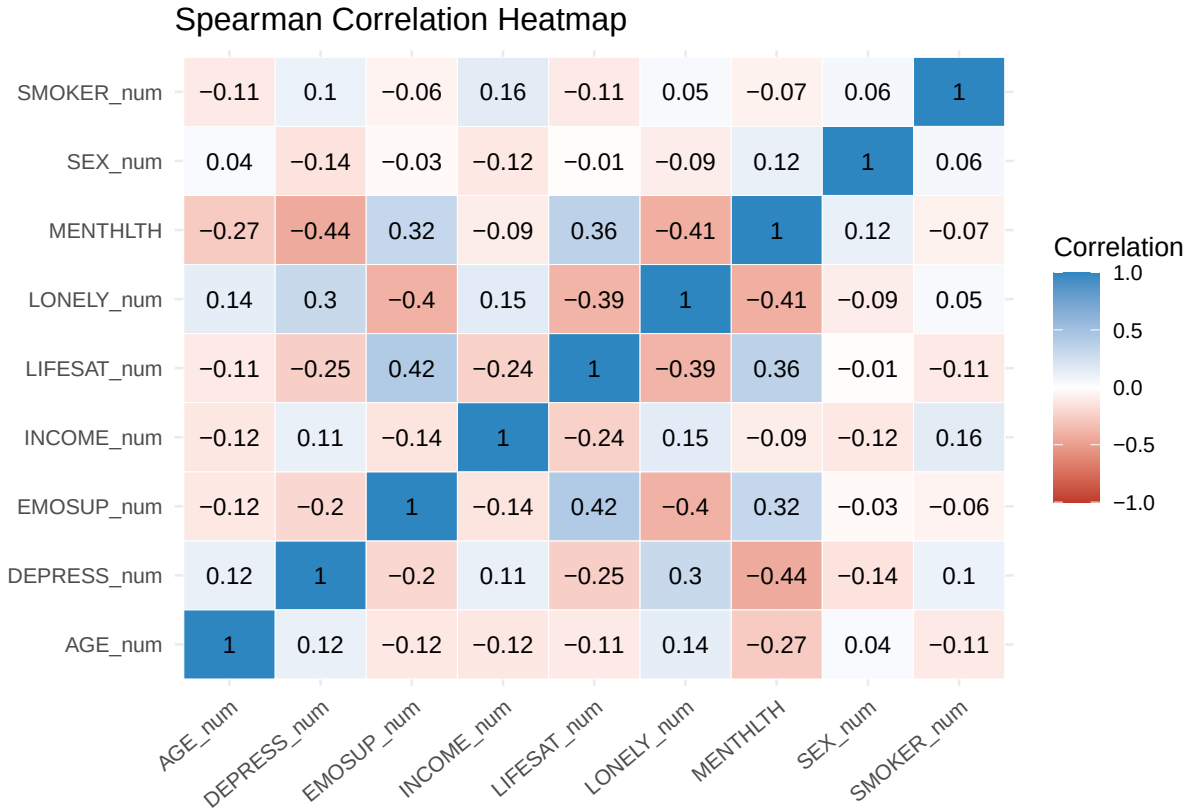
Table 10: Spearman Correlation with MENTHLTH (Mental Health Days)

	MENTHLTH
MENTHLTH	1.000
LONELY_num	-0.415
LIFESAT_num	0.360
DEPRESS_num	-0.436
EMOSUP_num	0.317
SEX_num	0.120
AGE_num	-0.266
INCOME_num	-0.093
SMOKER_num	-0.068

```
cor_long <- as.data.frame(cor_matrix) %>%
  tibble::rownames_to_column("Var1") %>%
```

```
  pivot_longer(-Var1, names_to = "Var2", values_to = "corr")

ggplot(cor_long, aes(x = Var1, y = Var2, fill = corr)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(corr, 2)), size = 3) +
  scale_fill_gradient2(low  = "#C0392B", mid = "white", high = "#2E86C1",
                       midpoint = 0, limits = c(-1, 1)) +
  labs(title = "Spearman Correlation Heatmap",
       x = "", y = "", fill = "Correlation") +
  theme_minimal(base_size = 10) +
  theme(axis.text.x = element_text(angle = 40, hjust = 1))
```



Interpretation: Loneliness and life satisfaction are expected to show the strongest correlations with mental health days – stronger than demographic factors like age and income – confirming that social determinants are critical drivers of mental health.

---

5. Multiple Linear Regression

```
reg_df <- inf_df %>%
  filter(!is.na(SEX), !is.na(AGE), !is.na(INCOME),
         !is.na(SMOKER), !is.na(LONELY), !is.na(EMOSUP),
         !is.na(LIFESAT), !is.na(DEPRESS)) %>%
  mutate(
    SEX_num     = as.numeric(SEX)     - 1,
    AGE_num     = as.numeric(AGE),
```

```
    INCOME_num  = as.numeric(INCOME),
    SMOKER_num  = as.numeric(SMOKER),
    LONELY_num  = as.numeric(LONELY),
    EMOSUP_num  = as.numeric(EMOSUP),
    LIFESAT_num = as.numeric(LIFESAT),
    DEPRESS_num = as.numeric(DEPRESS) - 1
  )

model <- lm(MENTHLTH ~ SEX_num + AGE_num + INCOME_num + SMOKER_num +
              LONELY_num + EMOSUP_num + LIFESAT_num + DEPRESS_num,
            data = reg_df)

tidy(model) %>%
  mutate(
    term = recode(term,
      SEX_num     = "Sex (Female=1)",
      AGE_num     = "Age Group",
      INCOME_num  = "Income Level",
      SMOKER_num  = "Smoking Status",
      LONELY_num  = "Loneliness",
      EMOSUP_num  = "Emotional Support",
      LIFESAT_num = "Life Satisfaction",
      DEPRESS_num = "Depression (Yes=1)"
    ),
    across(where(is.numeric), ~ round(., 4))
  ) %>%
  kable(caption = "Multiple Linear Regression: Predictors of Mental Health Days")
```

Table 11: Multiple Linear Regression: Predictors of Mental Health Days

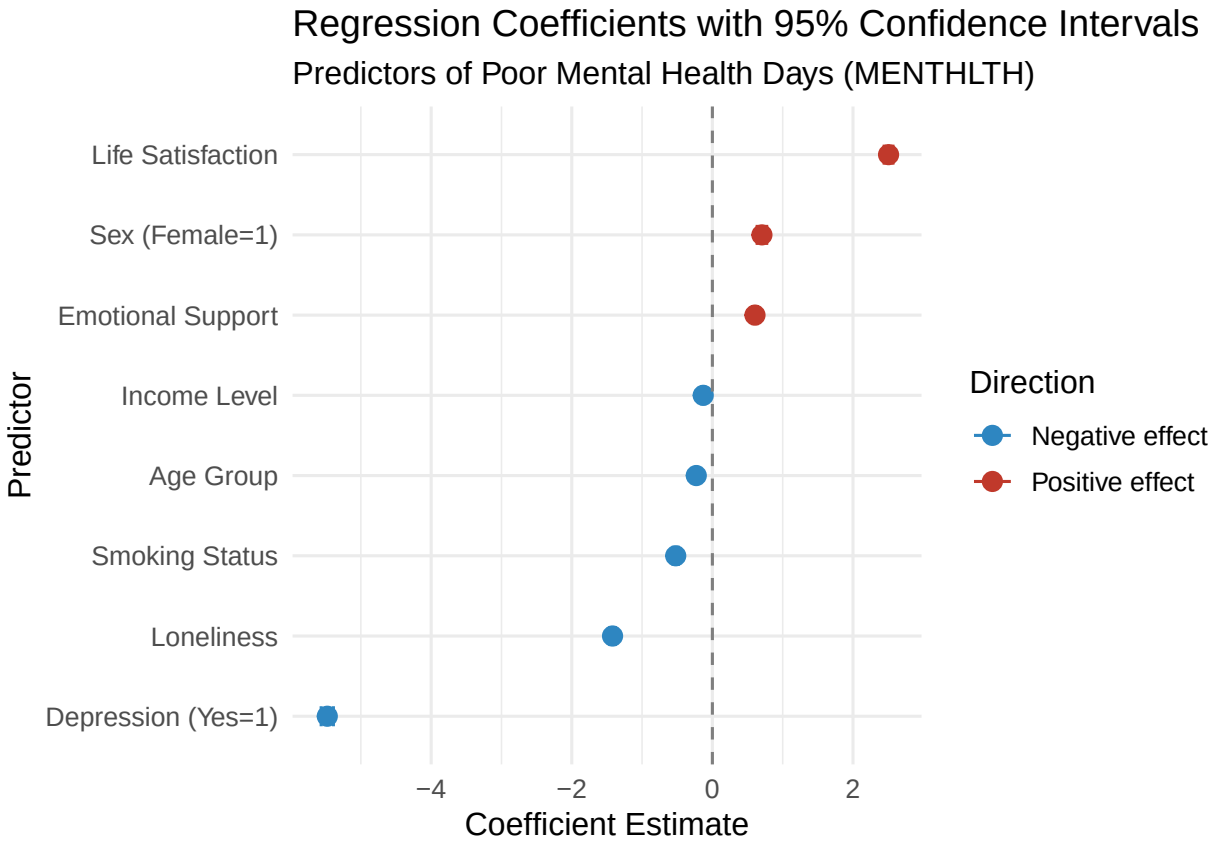
term	estimate	std.error	statistic	p.value
(Intercept)	12.9088	0.1524	84.7154	0
Sex (Female=1)	0.7047	0.0347	20.3225	0
Age Group	-0.2293	0.0049	-46.7340	0
Income Level	-0.1303	0.0113	-11.5364	0
Smoking Status	-0.5216	0.0199	-26.1626	0
Loneliness	-1.4200	0.0202	-70.2887	0
Emotional Support	0.6058	0.0182	33.3083	0
Life Satisfaction	2.5059	0.0319	78.5169	0
Depression (Yes=1)	-5.4776	0.0444	-123.4525	0

```
glance(model) %>%
  select(r.squared, adj.r.squared, statistic, p.value, df, nobs) %>%
  mutate(across(where(is.numeric), ~ round(., 4))) %>%
  kable(caption = "Model Summary Statistics")
```

Table 12: Model Summary Statistics

r.squared	adj.r.squared	statistic	p.value	df	nobs
0.3124	0.3124	9178.782	0	8	161616

```r
tidy(model) %>%
  filter(term != "(Intercept)") %>%
  mutate(
    lower = estimate - 1.96 * std.error,
    upper = estimate + 1.96 * std.error,
    term  = recode(term,
      SEX_num     = "Sex (Female=1)",
      AGE_num     = "Age Group",
      INCOME_num  = "Income Level",
      SMOKER_num  = "Smoking Status",
      LONELY_num  = "Loneliness",
      EMOSUP_num  = "Emotional Support",
      LIFESAT_num = "Life Satisfaction",
      DEPRESS_num = "Depression (Yes=1)"
    )
  ) %>%
  ggplot(aes(x = estimate, y = fct_reorder(term, estimate),
             color = estimate > 0)) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = lower, xmax = upper), height = 0.25) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  scale_color_manual(values = c("TRUE"  = "#C0392B",
                                "FALSE" = "#2E86C1"),
                     labels = c("Negative effect","Positive effect")) +
  labs(title    = "Regression Coefficients with 95% Confidence Intervals",
       subtitle = "Predictors of Poor Mental Health Days (MENTHLTH)",
       x = "Coefficient Estimate", y = "Predictor",
       color = "Direction") +
  theme_minimal(base_size = 12)
```



Interpretation:

- **Depression**: The largest positive coefficient – being diagnosed with depression adds the most poor mental health days, holding all else constant.
- **Loneliness**: Higher loneliness is associated with more poor mental health days – a key social determinant.
- **Life Satisfaction**: Lower satisfaction is associated with more poor mental health days.
- **Emotional Support**: Greater support is associated with fewer poor mental health days – confirming its protective effect.
- **Age**: Older age is associated with fewer poor mental health days.
- **R-squared**: With 8 predictors including social determinants, the model explains more variance than demographic-only models – showing the importance of social factors.

---

6. Summary of Key Statistical Findings

```
findings <- data.frame(
  Test = c(
    "t-Test: Sex vs MENTHLTH",
    "t-Test: Depression vs MENTHLTH",
    "ANOVA: Loneliness vs MENTHLTH",
    "ANOVA: Life Satisfaction vs MENTHLTH",
    "ANOVA: Emotional Support vs MENTHLTH",
    "Tukey HSD: Loneliness pairs",
    "Spearman Correlation",
    "Multiple Linear Regression (8 predictors)"
```

```
  ),
  Method = c(
    "Welch Two-Sample t-Test",
    "Welch Two-Sample t-Test",
    "One-Way ANOVA",
    "One-Way ANOVA",
    "One-Way ANOVA",
    "Tukey HSD Post-hoc",
    "Spearman Rank Correlation",
    "OLS Linear Regression"
  ),
  Key_Question = c(
    "Do males and females differ in mental health?",
    "Do depressed vs non-depressed differ?",
    "Does loneliness predict mental health?",
    "Does life satisfaction predict mental health?",
    "Does emotional support protect mental health?",
    "Which loneliness pairs are significantly different?",
    "How strongly are predictors related to mental health?",
    "What predicts mental health when controlling for all factors?"
  )
)
kable(findings, caption = "Summary of All Statistical Tests Performed")
```

Table 13: Summary of All Statistical Tests Performed

Test	Method	Key_Question
t-Test: Sex vs MENTHLTH	Welch Two-Sample t-Test	Do males and females differ in mental health?
t-Test: Depression vs MENTHLTH	Welch Two-Sample t-Test	Do depressed vs non-depressed differ?
ANOVA: Loneliness vs MENTHLTH	One-Way ANOVA	Does loneliness predict mental health?
ANOVA: Life Satisfaction vs MENTHLTH	One-Way ANOVA	Does life satisfaction predict mental health?
ANOVA: Emotional Support vs MENTHLTH	One-Way ANOVA	Does emotional support protect mental health?
Tukey HSD: Loneliness pairs	Tukey HSD Post-hoc	Which loneliness pairs are significantly different?
Spearman Correlation	Spearman Rank Correlation	How strongly are predictors related to mental health?
Multiple Linear Regression (8 predictors)	OLS Linear Regression	What predicts mental health when controlling for all factors?

All tests use a significance threshold of alpha = 0.05. Dataset: BRFSS 2024 – Social Determinant Module States (n = 200,326) Report generated using R 4.5.2